



Intel NUC mini-PC

Intel recently demonstrated its Next Unit of Computing PC which is based on a 4-in square motherboard, 4GB RAM and 40GB SSD



ZTE Blade III

ZTE will launch its new Blade III with Android 4.0 and 512MB RAM sometime later this year

Digit 101

Digit 101 - Motherboard

Your motherboard is not as boring as it may seem to be, there is some great technology that goes into making one. Here are some of the common components -

CPU Power

This is the region from where your CPU satisfies its power needs, the socket looks similar to a PCIe power connector but there are differences between them both with regards to the pinout. Check for the pins with corners cut to match the right cable. Variants - 4PIN, 6PIN, 8PIN, 8PIN+4PIN, 2x 8PIN/4PIN.

VRM circuitry

The power for the CPU is channelled via the Voltage Regulator Module. All the power phase designs that you get to read about is implemented in the VRM design.

RAM slots

Fits only one type - DDR1, DDR2 or DDR3. The notch and pin count helps to differentiate between these RAM types. Often colour coded to indicate which slots when occupied work in tandem to create a multi-channel mode. The max till now is 4-channels.

CPU Socket

This is where your CPU is installed. Always switch it on only if a CPU cooler is attached, they tend to get real hot real fast. This one is a LGA2011 socket.

CMOS Battery

It helps the CMOS to remember all settings. This is the reason why you don't have to set the date and time at every boot.

PCIExpress x16 slot

Offers the most bandwidth for peripherals. Often used by Graphics cards, RAID cards, Sound cards, PCI-e SSDs etc. Basically supports all PCI-e devices 1x/4x/8x/16x.

PCIExpress x1 slot

Commonly used by LAN cards and similar expansion cards.

PCI slot

An ancestor of the PCI-e slot that offers lesser bandwidth.

POST indicator

Power on self-test LED counter gives the error code if any occur during the boot sequence.

Front Panel Audio header

Audio I/O connector from your front panel goes here.

Power/Reset S/W

Convenient switches for folks who end up using the computer in an open configuration.

FireWire Header

If your cabinet has a front panel firewire connector then it'll be plugged into this 9 pin header. Also known as IEEE 1394.

Front panel switches/lights

This headers is where you hook up the Power s/w, Reset s/w, HDD LED, Power LED which are on the Front Panel of your computer cabinet.

CIR Header

If your cabinet has any Infrared transceivers then this is where you connect it to use it with your computer.

USB 2.0 Headers

If you feel the need for extra USB ports then connect the headers here. Each header gives you 2 USB ports but power is shared across them. Certain models come with high amperage headers which can charge demanding devices like iPods. Consists of 10-pins.

USB 3.0 Header

Connect your USB 3.0 cable here, also supports 2 ports off 1 header. Consists of 20 pins.

Motherboard Power

The entire motherboard is powered from this socket including certain low power peripherals like LAN cards, expansion cards etc. Often, if a peripheral needs more power then it will carry a separate connector on it, be it 4-pin Molex or SATA power.

PWM Fan header

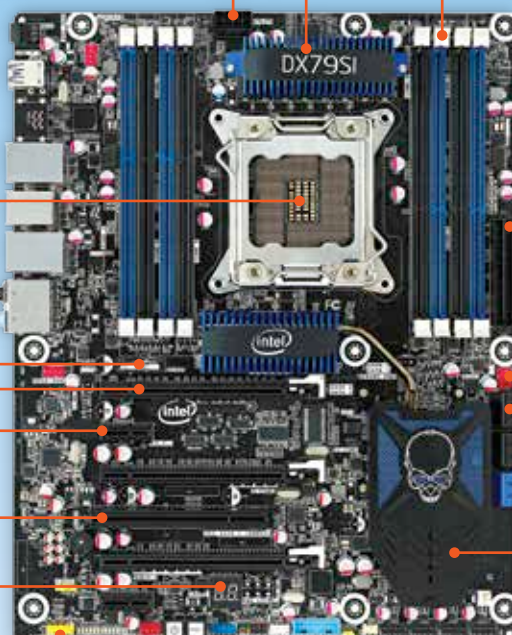
There'll be multiple such headers across the board. Connect your CPU fan to the one labelled for it to avoid boot-up issues. Can be 2-PIN, 3-PIN or 4-PIN. 3-PIN variants can sense the current RPM of your fan while 4-PIN variants are PWM so they can be better controlled.

SATA data sockets

Depending on the revision and make of the chipset (stock/third-party), you can connect SATA devices here. It's backwards compatible so you can connect old SATA I and II devices too. Certain third-party chipsets don't support ATAPI devices, so check your manual for more information.

PCH

Underneath this heatsink lies the Platform Controller Hub for the motherboard or simply put, the "chipset", it handles communication between all peripherals and the CPU.



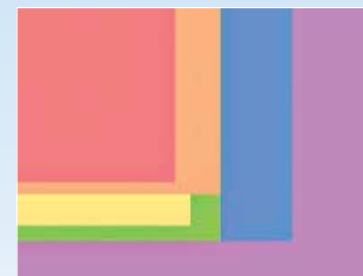
CPU sockets

1) Intel

Current - LGA 2011, 1155, 1156, 1366, 775
Future - LGA 1150

2) AMD

Current - AM2+, AM3, AM3+, FM1,
Future - FM2



ATX motherboard size comparison (in mm)

FlexATX.....	229x191
MicroATX/Embedded ATX.....	244x244
Mini ATX.....	284x208
Standard ATX.....	305x244
Extended ATX (EATX).....	305x330
Workstation ATX (WATX).....	356x425